

Robot Fish Simulator Survey

Six simulation tools applied to one soft, hydraulically actuated robot fish: what each one takes as input, what it produces, how it works inside, and what it can and cannot answer.

Branches: main · stonefish · pyelastica · sofa · openfoam · openmodelica Version: 9 Oct 2026
Language: EN (PL version available)

Read the numbers as mechanisms, not as a ranking. Every physical parameter in the six demos is a placeholder that has not been identified on a real robot. The demos also use different parameter sets (for example, the SOFA chamber holds 91 ml and the MuJoCo chamber 30 ml). A speed of 22 cm/s in one tool and 44 cm/s in another says which physics each tool includes, not which tool is more accurate.

1 · Summary

Each tool answers a different question about the same fish. Three tools simulate the whole fish swimming freely, two look closely at the soft tail, and one models the drive train from the battery to the fin.

TOOL	BRANCH / FOLDER	LEVEL OF DETAIL	WHAT IT SHOWS IN THIS REPO
● MuJoCo	main · MuJoCo/	Whole fish, rigid segments	Swimming, turning, depth control with a bladder; interactive keyboard control
● Stonefish	stonefish · Stonefish/	Whole fish, rigid segments, marine environment	Hydrodynamics from the mesh, VBS ballast, noisy pressure and IMU sensors, sea current
● PyElastica	pyelastica · fish_elastica_demo/	Whole fish as one soft rod	Chamber-driven bending, free 3D swimming, effect of stiffness on speed, depth control
● SOFA + SoftRobots	sofa · SOFA/	Tail only, 3D finite elements	How the silicone tail and its chambers deform: p–V curve, bend angle, flapping in air and water
● OpenFOAM + preCICE + CalculiX	openfoam · OpenFOAM/	Tail and water flow, 2D	Two-way fluid–structure interaction: vortex wake, added mass, tethered thrust
● OpenModelica	openmodelica · FishRobot/	System model, lumped	Battery → motor → pump → pipes → chambers → tail; energy budget, ballast, FMU export, calibration

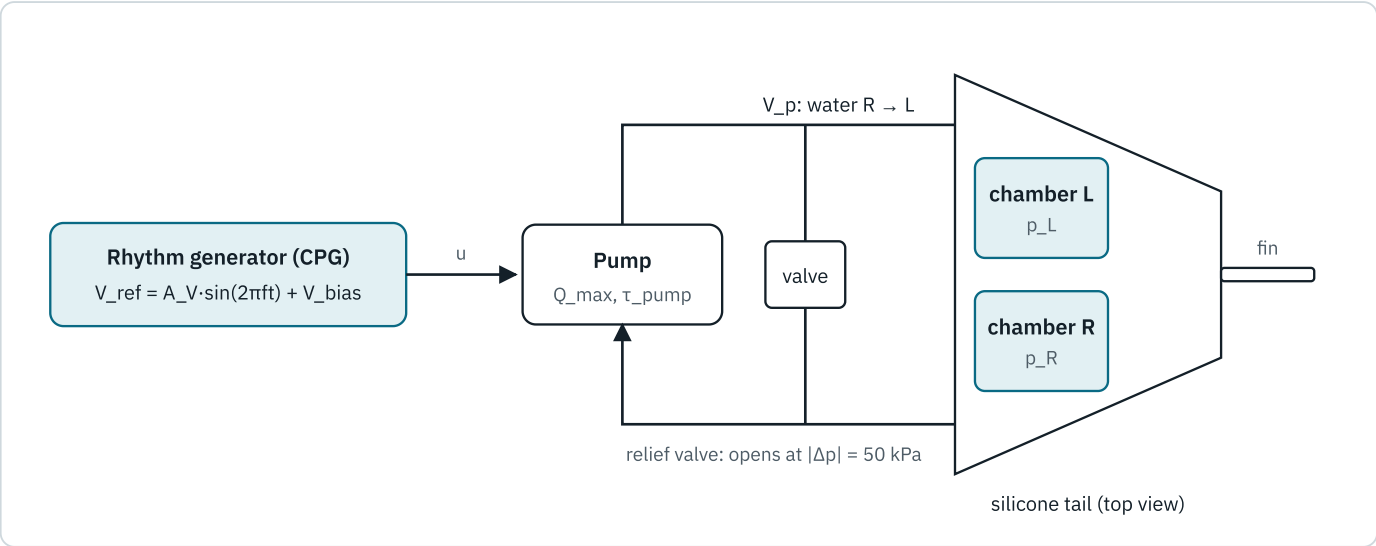
Which tool for which question

QUESTION	FIRST CHOICE	ALSO USEFUL
Does the fish swim, turn and hold depth? I want to tune controllers interactively.	● MuJoCo	Stonefish
Does the control software work with noisy sensors and a sea current? Path to ROS 2.	● Stonefish	MuJoCo
How does body and tail stiffness change speed, cheaply and with a continuous body?	● PyElastica	SOFA
What pressure does a chamber need for a given volume and bend? Which wall, fiber or spine design bends best?	● SOFA	OpenFOAM (CalculiX part)
What does the water really do: wake, added mass, real thrust or drag?	● OpenFOAM FSI	none of the others resolves flow

QUESTION	FIRST CHOICE	ALSO USEFUL
Are the motor, pump and battery sized right? How long does the battery last? Where is the energy lost?	● OpenModelica	none of the others has electrics
How do I turn bench measurements into model parameters?	● OpenModelica (calibrate.py)	calibration plans in the MuJoCo and SOFA READMEs

2 · The common fish model

All six demos model the same robot concept, a fish with a soft silicone tail bent by water pumped between two chambers. Knowing this loop makes the input tables in section 3 readable.



The shared hydraulic loop. The pump moves a volume V_p from chamber R to chamber L in a closed circuit; the pressure difference $\Delta p = p_L - p_R$ bends the tail. The rhythm generator commands a volume (not a flow), and an offset V_{bias} makes the fish turn. Depth is controlled separately: a PID sets the volume of a ballast bladder (MuJoCo, PyElastica), a VBS tank (Stonefish) or a syringe (OpenModelica).

The same quantities, set differently in each demo

QUANTITY	● MUJOCO	● STONEFISH	● PYELASTICA	● SOFA	● OPENFOAM	● OPENMODELICA
Body	ellipsoid hull 25×10×10 cm	same as MuJoCo	whole fish = rod 0.40 m	none (tail clamped)	2D head, half- ellipse 60 mm, fixed	lumped mass 1 kg
Tail	5 rigid segments × 40 mm, 3 actuated	5 rigid segments × 40 mm	continuous, chamber zone 0.20–0.32 m	3D FEM, 0.20 m	2D FEM, 0.15 m	1 rotational DOF
Pump flow Q_{max}	60 ml/s	60 ml/s	160 ml/s	250 ml/s	none (pressure prescribed)	0.3 ml per revolution
Volume amplitude A_V	8 ml	8 ml	8 ml	17 ml	$P_0 = 15$ kPa	CPG duty $A = 0.8$
Relief pressure	50 kPa	50 kPa	50 kPa	50 kPa	none	50 kPa
Ballast	bladder 0–40 ml	VBS 0.5–40.5 ml	bladder 0–40 ml	none	none	syringe 12.5 ml

3 · The six tools

Each tool section has the same parts: a fact sheet, the working principle, the inputs (what the demo uses and what the tool could also take), the outputs with example values from the repo, the demo stages, and the limitations.

● MuJoCo

A fast rigid-body engine with a simple built-in water model. The fish is a hull and a chain of five rigid tail segments; buoyancy and hydraulics are added in Python.

METHOD Rigid multibody dynamics + quasi-static ellipsoid fluid forces	INTERFACE Python bindings, MJCF model generated in code	TESTED VERSION MuJoCo 3.14.0, Python ≥ 3.10
INSTALL <code>pip install -r requirements.txt</code> , minutes	SPEED Live in the interactive viewer; scenarios of 10–130 s run headless	LICENCE* Apache-2.0

How it works

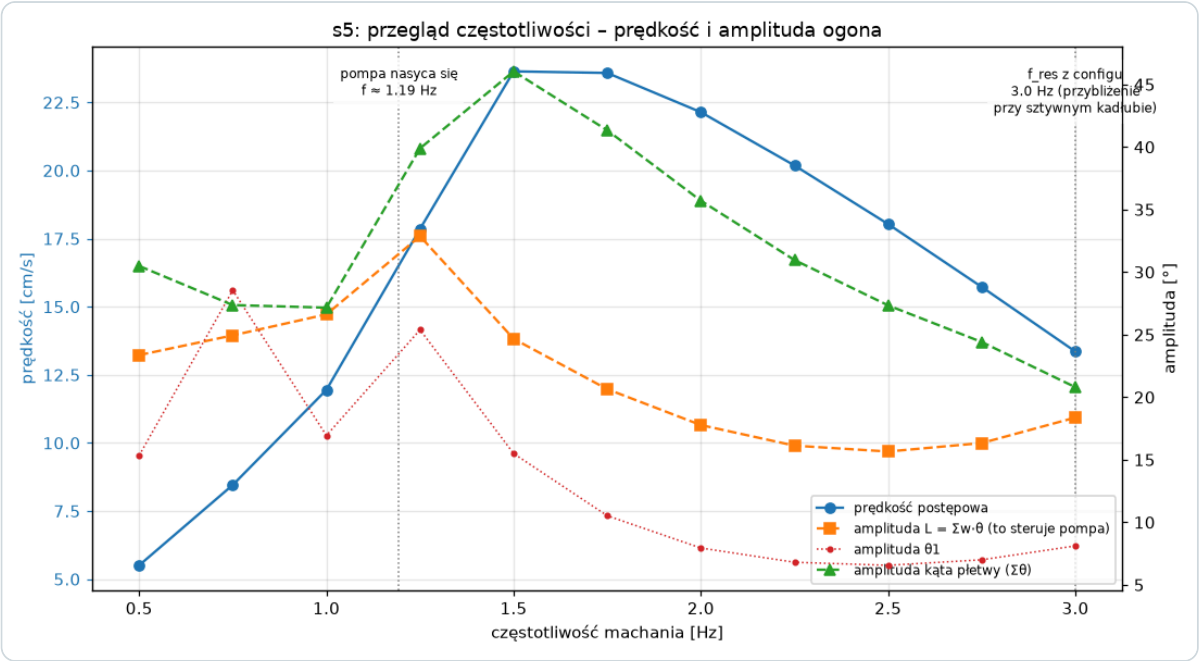
- **Body.** The hull is an ellipsoid on a free joint. The tail is five ellipsoids on hinge joints with springs and dampers, a pseudo-rigid-body approximation of the silicone. The caudal fin is a flat ellipsoid on the last segment.
- **Water.** MuJoCo's built-in ellipsoid model gives blunt and slender drag, angular drag, Kutta lift and Magnus force for each part. There is no flow field.
- **Added mass, corrected.** MuJoCo keeps only the gyroscopic part of the added-mass force. On its own that part pushed the fish backwards (4.5 cm/s instead of 22 cm/s). The demo switches it off and adds the added mass as joint armature instead.
- **Buoyancy.** MuJoCo does not compute buoyancy. The demo applies it to each body, with the centre of buoyancy fixed in the hull frame so that tilting produces a righting moment. The hull mass is solved for neutral buoyancy at a half-full bladder.
- **Hydraulics.** A Python model runs before every physics step: a first-order pump, $\Delta p = (V_p - A \cdot r \cdot L) / C_h$ clipped at ± 50 kPa, and a tendon force $F = A \cdot r \cdot \Delta p$ spread over the three actuated joints with weights from 1 to 0.4.
- **Control.** A volume rhythm generator for the tail and a cascade depth PID that sets the bladder volume, followed by a rate-limited ballast pump.

Inputs

CATEGORY	USED IN THE DEMO (FISHSIM/CONFIG.PY)	ALSO POSSIBLE WITH MUJOCO
Environment	water ρ 1000 kg/m ³ , μ 0.0009 Pa·s, step 2 ms (implicitfast), floor at -3 m	height fields, terrain meshes
Geometry	hull ellipsoid 0.25 × 0.10 × 0.10 m; 5 tail segments × 40 mm tapering to 0.4; fin 70 × 6 × 120 mm	MJCF or URDF files, STL/OBJ meshes
Joints	$\pm 35^\circ$, actuated stiffness 0.3 N·m/rad, passive stiffness from a 3 Hz resonance, damping 0.005 N·m·s/rad	any joint type, tendons, equality constraints
Water model	<code>fluidcoef</code> = blunt 0.5, slender 0.25, angular 1.5, Kutta 1.0, Magnus 1.0; added-mass mode	deformable flex bodies (the fluid model ignores them)
Hydraulics	A_{eff} 4·10 ⁻⁴ m ² , r_{eff} 0.05 m, C_h 8·10 ⁻¹¹ m ³ /Pa, p_{max} 50 kPa, chamber 30 ml, Q_{max} 60 ml/s, τ_{pump} 30 ms	any Python model coupled through <code>ctrl</code>
Ballast and control	bladder 0–40 ml at 10 ml/s; depth PID gains; tail 2 Hz, A_v 8 ml, V_{bias} , K_v 10 s ⁻¹	sensors (IMU, force, touch), cameras, GPU batches (MJX) for reinforcement learning
Live input	viewer keys: ←/→ $V_{\text{bias}} \pm 0.5$ ml, ↑/↓ frequency ± 0.25 Hz, PgUp/PgDn depth ± 0.25 m, Space start/stop	any Python callback

Outputs

OUTPUT	CONTENT	EXAMPLE VALUE
s1 straight swim	CSV + PNG: speed, five joint angles, p_L, p_R, Δp, tendon force, pump command	≈22 cm/s at 2 Hz after ≈2 s
s2 water on/off	speed with water, without water, with MuJoCo's default added mass	22 / 0 / 4.5 cm/s
s3 turning	XY trajectories for V_bias from −3 to +3 ml	±3 ml: 6.3°/s, radius ≈1.6 m
s4 depth steps	depth, setpoint, bladder volume, pitch	1 m in ≈35 s swimming, ≈6 s hovering; overshoot ≈3 cm
s5 frequency sweep	speed, amplitudes, pump saturation for 0.5–3 Hz	peak 23.6 cm/s at 1.5 Hz
s6 righting	roll, pitch and depth after a 30° roll	upright in ≈0.3 s
Model report	text: masses, volumes, net buoyancy, CG/CB, stability margins of the time step	hover drift 0.000 cm in 5 s



Stage 6 frequency sweep (plot labels are in Polish on the main branch). Speed peaks at 1.5 Hz, just above 1.19 Hz, where the pump starts to saturate.

Demo stages

- 0
- check_api.py: checks the API; a ball with the density of water sinks, which shows that MuJoCo has no built-in buoyancy
- 1
- show_model.py: static model and mass/buoyancy report
- 2–5
- run_scenarios.py: hover and righting, offline hydraulics, swimming with the added-mass diagnosis, turning, depth steps
- 6
- sweep_frequency.py: frequency sweep
- 7
- run_viewer.py: interactive control from the keyboard

Limitations

- No flow simulation: no wake, no vortices, no interaction between body parts through the water.
- Diagonal added mass, computed once for a straight tail, so the fish accelerates too quickly.
- The tail is rigid segments, not a continuum. The hydraulics are linear (constant C_h, ideal valve, no losses).
- No waves or free surface; buoyancy is faded out over ±5 cm near the surface as a patch.

● Stonefish

A marine-robotics simulator that computes water forces from the actual surface mesh and ships sensors, ballast tanks and ocean currents. The fish has the same layout as in MuJoCo; the extra physics is written in C++.

METHOD Bullet multibody (Featherstone) + per-triangle hydrodynamics	INTERFACE C++20 application + XML scenarios + JSON config	TESTED VERSION Stonefish 1.5.0, built from source
INSTALL CMake build; OpenGL ≥ 4.3, SDL2, Freetype, GLM, OpenMP	SPEED 5–8× faster than real time in console mode at 2000 Hz; GUI ≈800 FPS	LICENCE* GPL-3.0

How it works

- **Body.** Head plus five tail links on revolute joints and a fin. The meshes are generated by `make_meshes.py` (320 triangles for physics, 5120 for display), which also balances the mass and sizes the steel ballast.
- **Water from geometry.** Buoyancy is computed from hydrostatic pressure on every triangle. Pressure drag acts per face, skin friction acts along the surface, and added mass comes from a fitted ellipsoid.
- **Fixes in C++.** Face drag cannot produce lift, so `FinLift.cpp` adds a flat-plate lift on the fin. Without it the fish reaches only 2.7 cm/s instead of 34 cm/s. The library's skin friction was about 250× too high and is replaced by a Blasius law.
- **Drive.** `Hydraulics.cpp` holds the same lumped pump and chamber model as MuJoCo. `TailDriver.cpp` applies joint torques $\tau_i = w_i F - k_i \cdot \theta_i$ as an internal drive (action equals reaction).
- **Sensors and control.** The pressure sensor (50 Hz, σ 50 Pa) and the IMU (100 Hz) are noisy. The depth PID reads the filtered pressure and sets the VBS water volume; a heading PI on the IMU yaw sets V_{bias} .
- **Ocean.** Seabed, posts for visual reference, water type and a uniform current.

Inputs

CATEGORY	USED IN THE DEMO	ALSO POSSIBLE WITH STONEFISH
Config file	<code>config/default.json</code> , per-scenario patches <code>sN_*.json</code> , single overrides <code>--set key=value</code> (no recompilation)	any parameter exposed by the application
Scenario XML	<code>fish_base.scn</code> : links, joints, motors, VBS, pressure sensor, IMU, encoders; <code>world_base.scn</code> : seabed, ocean, current	waves (GPU only), other current types, custom hydrodynamic coefficients
Geometry and materials	OBJ meshes; hull density 600, silicone 1100, steel ballast 7850 kg/m³ (sized automatically)	any OBJ geometry
Hydraulics and rhythm	same values as MuJoCo: Q_{max} 60 ml/s, A_V 8 ml, f 2 Hz, p_{max} 50 kPa; fin lift slope 2.8/rad	thrusters, rudders, servos
Sensors	pressure 50 Hz with σ 50 Pa; IMU 100 Hz with angle, gyro and accelerometer noise	cameras, sonar, DVL, GPS, thermal camera
Control and live input	depth setpoint schedule, heading PI; GUI keys for the CPG, V_{bias} , frequency and depth	ROS 2 interface (<code>stonefish_ros2</code>), not used here
Simulation	2000 steps/s (2 ms was unstable), hydrodynamics every step, log every 20 steps	real-time or batch mode

Outputs

OUTPUT	CONTENT	EXAMPLE VALUE
CSV log, 100 Hz	≈70 columns: pose, velocities, sensor readings, joint $\theta/\omega/\tau$, pump, chambers, VBS, forces per body part	full logs in <code>results/logs/</code>
s1 hover / righting	depth drift; roll after a 30° tilt	0.21 mm in 10 s; 30° → 10° in 5 s
s2 swim	speed with and without fin lift, locked tail; force balance	34 cm/s; 2.7 cm/s without lift; fin lift 0.51 N
s3 turning	turning radius for $V_{\text{bias}} \pm 1/\pm 2/\pm 3$ ml	20.4 / 10.1 / 6.6 m
s4 depth	true vs measured depth with the noisy sensor	1 → 3 m: overshoot 3.3 cm, within ± 5 cm after 21 s
s5 current	drift in a 5 cm/s side current, with and without heading control	heading held at +2.9°, yet 1 m sideways drift in 30 s
s6 sweep, GUI	speed vs frequency; summary . json; 3D window	peak 33.9 cm/s at 2 Hz



The Stonefish GUI during scenario s2: the panel on the right shows live speed, true and measured depth, chamber pressures and VBS volume.

Demo stages

- 1–2 Build Stonefish; generate meshes and the mass balance (`tools/make_meshes.py`)
- 3 s1: hovering and righting
- 4 s2: joints and hydraulics, diagnosing where thrust comes from
- 5 s3–s5: turning, depth on the noisy sensor, side current
- 6 s6: frequency sweep; `t_internal` checks that the drive is internal
- 7 ROS 2: skipped

Limitations

- Rigid-segment tail; quasi-static hydrodynamics with no wake, no delayed stall and no interaction between bodies.
- Added mass averaged over the axes, with no added inertia about the roll axis; friction linearised at 0.2 m/s.
- The GUI needs a GPU. Eight library pitfalls of version 1.5 are documented together with their workarounds.

● PyElastica

The whole fish is one soft Cosserat rod: a centreline with a rotating frame that can bend, twist, shear and stretch. It is the cheapest way here to simulate a continuously deforming body that swims freely.

METHOD Cosserat rod theory, explicit symplectic Position Verlet	INTERFACE Pure Python, <code>import elastica as ea</code> , numba kernels	TESTED VERSION PyElastica 1.0.0, numba 0.68, Python 3.14
INSTALL <code>pip install -r requirements.txt</code> , minutes	SPEED ≈27 μs per step of a 100-element rod; 9–70 s per stage	LICENCE* MIT

How it works

- **Body.** A 0.40 m rod with 50 elements. Each element has an elliptical section (from a table of 7 stations), so the tail is soft sideways and stiff vertically. The head is 20× stiffer.
- **Actuation from inside.** The chamber pressure changes the rest curvature of the chamber zone (0.20–0.32 m): $\kappa_{\text{rest}} = -A_r \cdot \Delta p / B$. The drive produces no net force; the centre of mass drifts only 10^{-15} m in vacuum.
- **Hydraulics.** The same lumped pump, chamber and valve model as in MuJoCo and SOFA, stepped every 0.2 ms and coupled through the bend angle of the chamber zone.
- **Water.** A custom force: quadratic Morison drag per unit length, plus the Lighthill large-amplitude reactive force at the trailing edge. The built-in Stokes slender-body model is kept only to show it is wrong at this Reynolds number ($\approx 1.8 \cdot 10^5$): its force is ≈400× too small.
- **Buoyancy and depth.** Per-element weight and buoyancy, a righting moment from the centre of gravity sitting below the axis, a bladder, and a cascade depth PID ported from MuJoCo.
- **Time step.** ≈19 μs, from the explicit stability limit of the rod.

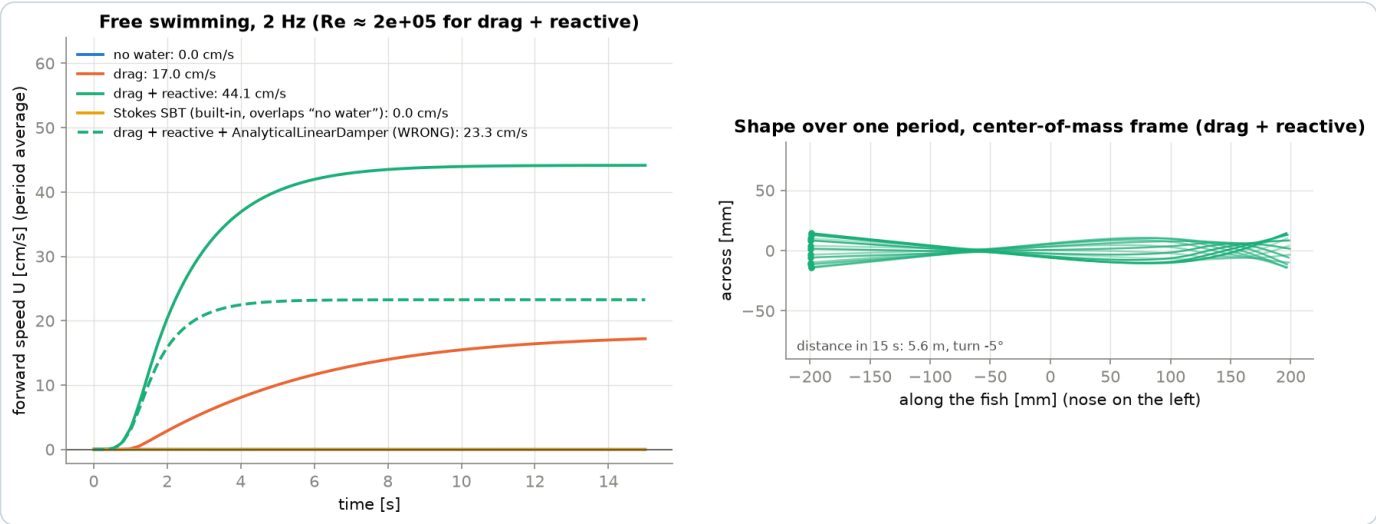
Inputs

CATEGORY	USED IN THE DEMO (FISHROD/CONFIG.PY)	ALSO POSSIBLE WITH PYELASTICA
Rod	L 0.40 m, ρ 1050 kg/m ³ , E 0.5 MPa, ν 0.5, 50 elements	several rods joined by joints, rigid bodies
Shape	half-width and half-height at 7 stations (24 × 34 mm at mid-body), ellipse or circle, stiff head	any per-element radius and stiffness
Hydraulics	A_{eff} 4.3·10 ⁻⁴ m ² , r_{eff} 5.5 mm, C_h 4.5·10 ⁻⁹ m ³ /Pa, p_{max} 50 kPa, Q_{max} 160 ml/s, f 2 Hz, A_V 8 ml	built-in muscle torques, uniform torques
Water	model: none, drag, drag+reactive, stokes_sbt; C_n 1.2, C_t 0.01	coupling to an immersed-boundary flow solver (SophT)
Buoyancy and control	CG 5 mm below the axis, bladder 0–40 ml at 10 ml/s, PID at 100 Hz, depth schedule	contact and friction with a floor
Stage settings	sweep grids (f 0.5–3 Hz × E ×0.5/1/2), durations, damping type	other boundary conditions and dampers

Outputs

OUTPUT	CONTENT	EXAMPLE VALUE
e1 validation	cantilever vs theory, natural frequency, energy drift for $n = 25/50/100$	$n = 50$: tip error 1.05 %, ω_1 error 1.8 %
e2 actuation	bend vs pressure and vs volume; internal-drive check	8 ml → 15.8 kPa, 0.377 rad

OUTPUT	CONTENT	EXAMPLE VALUE
e3 tethered	tip amplitude, phase, thrust and lateral force for each water model	thrust 149 mN with the reactive force, 5.9 mN with drag only
e4 free swim	speed vs time; body shape over one period; free_swim.gif	44.1 cm/s (1.1 body lengths/s); 17 cm/s with drag only
e5 sweep $f \times E$	speed, Strouhal number, tail natural frequency	softest tail at 3 Hz: 55 cm/s; f_n 1.89/2.60/3.64 Hz
e6 depth	depth, bladder volume, pitch, roll, speed	1.0 $\rightarrow$ 1.5 m settles in 9.9 s, overshoot 23 cm



Stage 4: the choice of water model decides the result. Without the reactive (Lighthill) force the fish reaches 17 cm/s; the Stokes model gives zero.

Demo stages

- 1 Rod validation against beam theory, convergence in $1/n$
- 2 Rest-curvature actuation and hydraulic coupling; internal-drive check in vacuum
- 3 Tethered tail with three water models; thrust from momentum balance
- 4 Free planar swimming, comparison of water models and damping
- 5 Frequency $\times$ stiffness sweep, resonance, Strouhal number
- 6 3D with gravity, buoyancy, bladder and depth PID while swimming
- 7 Stiffness export to MuJoCo: not done

Limitations

- Local water model with no wake: the Strouhal number is too low (0.12–0.2) and speed is overestimated.
- No distributed added mass, so the natural frequency in water is too high. The section is an ellipse, not a thin fin plate.
- The chamber is reduced to $A_r \cdot \Delta p$ and C_h ; elasticity is linear. A very stiff hydraulic spring breaks the explicit coupling.

● SOFA + SoftRobots

A 3D finite-element model of the silicone tail with real chamber cavities. The pump prescribes the chamber volume and SOFA computes the pressure needed. The tail is clamped at the root, as on a thrust test rig.

METHOD Corotational FEM on tetrahedra + Lagrange constraints for chambers	INTERFACE SofaPython3 scenes (createScene), C++ engine	TESTED VERSION SOFA v26.06.00 binary (SoftRobots included)
INSTALL scripts/setup.sh: ≈2 min + 230 MB, builds the CHOLMOD plugin	SPEED 100–800× slower than real time (coarse mesh, one core)	LICENCE* LGPL-2.1+ (SOFA), LGPL-3.0 (SoftRobots)

How it works

- **Mesh.** gmsh builds the tail with two cavities. The coarse level has 28 112 tetrahedra and 7 744 nodes; the fine level has 142 096 tetrahedra.
- **Material.** Corotational linear elasticity (large rotations, small strains), E 0.3 MPa, ν 0.45. The default design V4 adds a spine 20× stiffer in the septum and tension-only hoop fibers every 4 mm.
- **Chambers.** SurfacePressureConstraint in volume-growth mode: the pump sets the volume, and the constraint solver returns the pressure.
- **Solver.** Implicit Euler, a Gauss–Seidel constraint solver and the CHOLMOD sparse Cholesky factorisation (4–15× faster than LDL). Static curves are found pseudo-statically, holding each point until the motion dies out.
- **Water.** Custom quadratic normal drag and tangential friction on every skin triangle, applied explicitly; the silicone weight is reduced by buoyancy. There is no added mass, lift or wake.
- **Hydraulics.** A port of the MuJoCo rhythm with pump-lag compensation (without it the volume overshoots by 16 % at 2 Hz), a closed circuit and a 20 ml prefill.

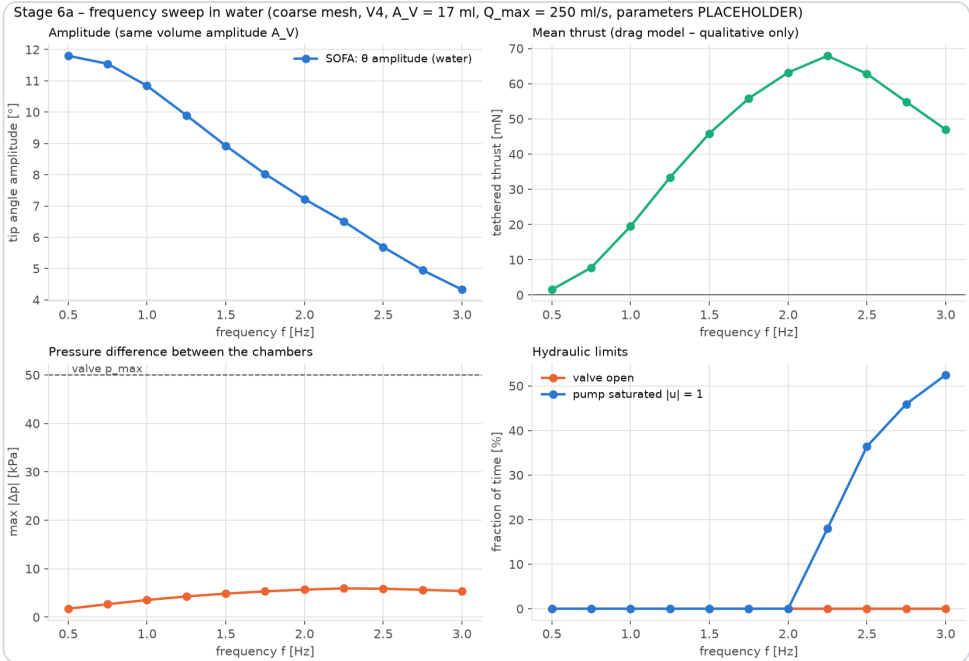
Inputs

CATEGORY	USED IN THE DEMO (FISHSOFA/CONFIG.PY)	ALSO POSSIBLE WITH SOFA
Geometry	tail 0.20 m, section 60 × 80 mm tapering to 0.4, fin 70 × 120 × 6 mm; two chambers 0.12 m long, 4 mm wall, 4 mm septum (91 ml each)	any Gmsh, STL, OBJ or VTK mesh; XML .scn scenes
Material	E 3·10 ⁵ Pa, ν 0.45, p 1100; spine E × 20; fibers 2·10 ⁵ N/m every 4 mm	hyperelastic materials, beams (BeamAdapter)
Actuation	f 2 Hz, A_V 17 ml, prefill 20 ml, Q_max 250 ml/s, p_max 50 kPa	pressure control, cable actuators, inverse control (SoftRobots.Inverse)
Environment	air or water; drag coefficients C_n 1.0, C_t 0.01	contact and collision
Numerics	step 2 ms (0.5–1 ms in water), Rayleigh damping, solver cholmod/ldl/cg/warp, mesh level test/coarse/medium/fine	model order reduction (up to ≈50× faster)
Run control	environment variables FISHSOFA_LEVEL, _MODE, _ENV, _WORKERS	GUI interaction

Outputs

OUTPUT	CONTENT	EXAMPLE VALUE
s1 mesh and sag	mesh quality report; tip sag under own weight on three meshes	−40.5 mm (coarse) to −41.2 mm (fine)
s2a design variants	bend for homogeneous, spine, thick wall, fibers, spine + fibers	1.3° homogeneous vs 31° for V4
s2 p–V curve	pressure and tip angle vs chamber volume, mesh convergence	50 ml: 16.1 kPa and 15.4° (fine mesh)
s3 symmetry	left vs right chamber	error ≤ 0.75 % in angle

OUTPUT	CONTENT	EXAMPLE VALUE
s4 flapping in air	angle, pressures, volume conservation, solver iterations	$\pm 12.7^\circ$ at 2 Hz, $ \Delta p \leq 5.7$ kPa
s5 water vs air	amplitude, phase, tethered thrust, lateral force, drag power	$\pm 7.3^\circ$ in water (0.55× air); thrust 66.7 mN
s6 sweeps, video	frequency and Young's modulus sweeps; flapping.mp4 1080p60	max thrust 67.9 mN at 2.25 Hz



Stage 6 in water. Amplitude falls steadily with frequency (no resonance in water), thrust peaks at 2.25 Hz, and above 2 Hz the pump saturates. The thrust comes from a drag model and is qualitative.

Demo stages

- 0 Component check; meaning, sign and units of the chamber volume constraint (≈ 2.5 min)
- 1 Meshes and sag under own weight (≈ 10 min)
- 2 Design variants V0–V4; p–V and angle curves, mesh convergence (≈ 35 min)
- 3 Left/right symmetry (≈ 3 min)
- 4 Antagonistic flapping in air (≈ 25 min)
- 5 Water vs air, tethered thrust (30–60 min)
- 6 Frequency and stiffness sweeps, 19 runs (≈ 1 h on 12 threads)

Limitations

- No flow and no reactive force, so the thrust is qualitative. The tail is clamped; the fish does not swim.
- Linear tetrahedra lock in thin walls, which overestimates pressure by 15–25 %. The material is corotational linear, not hyperelastic.
- Slow: explicit drag limits the step to 0.5 ms in water. A bug in SOFA v26.12-dev keeps the demo on v26.06.
- Exporting equivalent joint stiffnesses to MuJoCo (stage 7) was not done.

● OpenFOAM + preCICE + CalculiX

The only demo that solves the actual water flow. A CFD solver and a structural FEM solver exchange forces and displacements every time step until they agree. It is a 2D slice of the tail with the head held fixed.

METHOD Partitioned FSI: finite volumes + nonlinear FEM, implicit coupling	INTERFACE OpenFOAM dictionaries, CalculiX .inp, preCICE XML; Bash and Python tools	TESTED VERSIONS OpenFOAM v2606, CalculiX 2.20, preCICE 3.4.1
INSTALL setup.sh, ≈1.5 h, mostly built from source	SPEED 0.6–1 s per 2.5 ms window; stages 4–6 take 3 h 47 min	LICENCE* GPL-3.0 (OpenFOAM), GPL-2.0+ (CalculiX), LGPL-3.0 (preCICE)

How it works

- **Fluid.** pimpleFoam solves incompressible laminar Navier–Stokes in 2D (one cell deep, so forces are per metre of span) on a mesh that deforms with the tail.
- **Solid.** CalculiX runs dynamic nonlinear FEM (large deformations, linear elastic silicone, E 0.3 MPa) of a tail with PneuNet-style chambers.
- **Coupling.** preCICE maps fluid forces onto the solid and solid displacements onto the fluid mesh, and repeats each 2.5 ms window with the IQN-ILS quasi-Newton method until both agree (4.7 iterations on average).
- **Why implicit.** Silicone is almost as dense as water, so explicit coupling is unstable for any time step (the added-mass effect). Stage 3 shows the force growing 13× per window.
- **Actuation.** Prescribed chamber pressure $p_L = P_0 \cdot \max(0, \sin 2\pi ft)$, with the opposite half-wave in p_R . Pressure is the input, not volume.
- **Thrust.** The cycle-averaged axial force on the fixed body (head + tail).

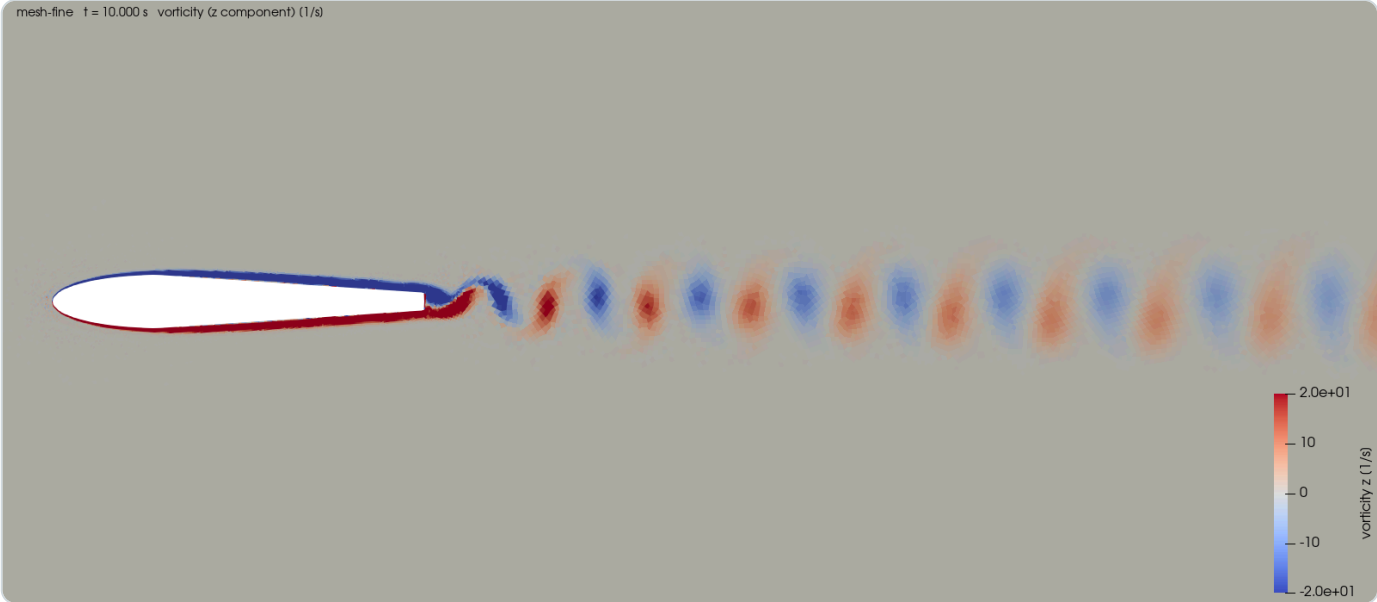
Inputs

CATEGORY	USED IN THE DEMO	ALSO POSSIBLE WITH THIS TOOLCHAIN
Geometry (params.py)	tail 0.15 m, thickness 30 → 10 mm, chambers at 10–110 mm, 3 mm walls, 10 cells per side; head half-ellipse 60 mm; domain 3L upstream, 8L downstream	full 3D geometry
Materials and water	$E \cdot 3 \cdot 10^5$ Pa, ν 0.45, ρ 1070 kg/m ³ ; water ρ 1000, ν 10 ^{−6} m ² /s	hyperelastic material, material damping (CalculiX)
Case (new-fsi-case.sh)	inflow speed U, time step, end time, P0, frequency, implicit/explicit coupling	free body with a 6-DOF solver
OpenFOAM setup	PIMPLE + GAMG, Laplacian mesh motion, force and vorticity function objects, inlet ramp, slip sides	RANS/LES turbulence, overset or remeshing for large amplitude
CalculiX setup	fsi.inp: nonlinear dynamics, HHT α −0.2, PARDISO solver, pressure amplitude tables	contact between chamber walls
preCICE setup	RBF mapping (5 mm radius), IQN-ILS, convergence 10 ^{−3} , tip watch point	more participants, e.g. a volume-driven hydraulics model
Mesh	gmsh, refinement via MESH_SCALE (0.7 / 1.0 / 1.4)	any OpenFOAM mesh generator

Outputs

OUTPUT	CONTENT	EXAMPLE VALUE
Flow fields	velocity, pressure, vorticity, time averages for ParaView	Kármán vortex street behind the body

OUTPUT	CONTENT	EXAMPLE VALUE
Forces	pressure and viscous force and moment per patch (head, tail)	rigid body at 0.2 m/s: drag 0.13 N/m, $\frac{2}{3}$ pressure
E1 dry tail	tip deflection vs pressure, L/R symmetry	1.2 mm/kPa; 24 mm (16 % of L) at 20 kPa
E3 coupling	explicit vs implicit stability	explicit diverges; implicit stable
E4 actuated	tip amplitude in water vs dry, thrust per cycle	28.3 mm in water vs 20 mm dry at 1 Hz
E5 inflow	mean axial force and Strouhal number vs speed	net drag at 0.05–0.2 m/s; zero only near 0.02 m/s
E6 frequency	amplitude and thrust vs frequency	resonance in water ≈ 1 Hz (dry 3.39 Hz)



Stage 2, fine mesh, $U = 0.2$ m/s: the vortex wake that none of the other five tools can show.

Demo stages

- 0 Official perpendicular-flap tutorial, to validate the install (34 s)
- 1 Solid only: deflection vs pressure (≈ 10 s)
- 2 Fluid only, rigid body: drag and mesh convergence (7 min)
- 3 Passive tail: explicit vs implicit coupling (10 min)
- 4 Actuated tail in still water vs dry (16–80 min)
- 5 Inflow at 0.05/0.1/0.2 m/s (≈ 30 min each)
- 6 Frequency sweep 0.5–2 Hz (16–102 min each)

Limitations

- 2D: no tip vortices, and forces are per metre of span. Laminar flow at $Re \approx 4 \cdot 10^4$.
- Fixed head, so no free swimming or recoil. Actuation is by pressure, not volume. No material damping.
- Mesh deformation limits the tip amplitude to ≈ 15 % of L. Two uniform chambers make a standing wave and a blunt tip, so the tail produces net drag.
- Slow and the most complex of the six to install.

● OpenModelica

An equation-based model of the whole drive train, from the battery through the motor, pump, pipes and chambers to a one-degree-of-freedom tail. It is the only demo with electrics, an energy budget, FMU export and a full calibration workflow.

METHOD Acausal Modelica equations → DAE → C code, variable-step solver	INTERFACE Modelica (OMEdit GUI), Python through OMPython and FMPy	TESTED VERSION OpenModelica 1.27.1, MSL 4.1.0
INSTALL apt repository (install_system.sh) + venv (install_user.sh)	SPEED 16 simulations ≈0.2 s after ≈1 s of compilation	LICENCE* OSMC-PL / GPL-3 (OpenModelica), BSD-3 (MSL)

How it works

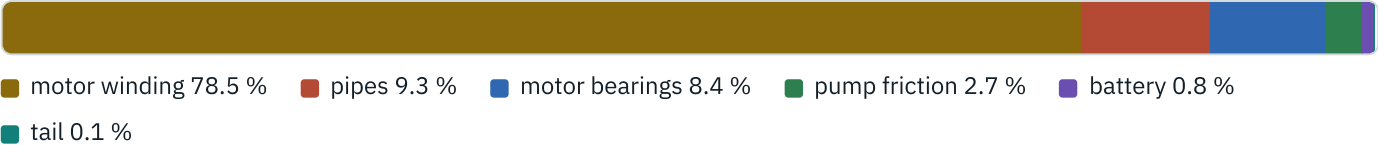
- **Acausal components.** Parts are joined by physical connectors. The custom hydraulic port carries pressure p and flow V_{flow} , whose product is power. The compiler sorts the equations itself, so there is no input/output wiring.
- **Drive chain.** 2S LiPo battery 7.4 V → H-bridge → DC motor → gear pump ($Q = D \cdot \omega - k_{\text{leak}} \cdot \Delta p$) → pipes (laminar plus Haaland turbulent friction, minor losses) → chambers with a nonlinear p - V table → relief valves.
- **Tail and fin.** The tail is a rotary piston with one degree of freedom: $(J + J_{\text{added}}) \cdot \ddot{\theta} = \tau - k\theta - c\dot{\theta} - c_h |\dot{\theta}| \dot{\theta}$. Thrust comes from Lighthill's rigid-fin formula and drives a surge equation.
- **Ballast.** A motor and lead screw drive a syringe piston; a heave equation and a cascade depth PID close the loop.
- **Energy.** Every loss is integrated, and the balance closes to $\approx 10^{-6}$.
- **Export and calibration.** TailDrive exports as an FMU 2.0 for co-simulation. Ten bench models and `calibrate.py` (least squares) identify the parameters step by step, from the motor to the water.

Inputs

CATEGORY	USED IN THE DEMO	ALSO POSSIBLE WITH MODELICA
Electrical	battery 7.4 V, R_{int} 0.05 Ω , 16.3 Wh; motor R 1 Ω , L 0.5 mH, k 0.01 N·m/A, rotor inertia	SoC-dependent voltage, PWM losses, thermal models
Hydraulic	pump 0.3 ml/rev, leakage, η 0.8; pipe 0.2 m × 4 mm, ζ 1.5, roughness 25 μm ; valve 50 kPa; chamber p - V table (11 points or a CSV file)	Modelica.Fluid and media, compressible or two-phase (a variant is already in HydraulicsFluid/)
Tail and fin	displacement $2 \cdot 10^{-5}$ m ³ /rad, k 2 N·m/rad, inertia, added inertia, damping; fin span 0.06 m, C_T 1.0	a p - V curve from SOFA or from measurement
Body and ballast	m 1 kg, added mass, C_d 0.3, area 0.005 m ² ; syringe 20 mm × 40 mm (12.5 ml), lead 1 mm	multibody library for 3D motion
Control	CPG amplitude 0.8, 1.5 Hz, bias; depth PID gains	any block-diagram controller
Measured data	CSV files for <code>calibrate.py</code> : motor step, pump, pipe, chamber cycles, valve, tail, thrust, hull tow and coast-down, ballast	tables from .mat/.txt, FMU import
Run time	-override parameters, OMEdit modifiers	solver choice: DASSL, IDA, CVODE, Euler

Outputs

OUTPUT	CONTENT	EXAMPLE VALUE
Time series (.mat)	every variable: p_L, p_R, flows, currents, motor speed, θ , thrust, speed, depth, ballast volume	valve opens at ≈ 51 kPa (HydraulicsStep)
Energy budget	loss per component, mechanical output, stored energy, balance error	1.8 W at 1 Hz $\rightarrow \approx 9.2$ h on 16.3 Wh
Swimming	surge speed, thrust, battery power vs frequency	best 8.5 cm/s at 0.5 Hz; 16.6 W at 4 Hz
Depth control	depth, piston position, current	3 cm/s descent, 6 % overshoot, 5 mm error
FMU	TailDrive.fmu (linux64, 1.8 MB) + comparison with OpenModelica	difference 0.016 % of amplitude
Calibration reports	fitted parameters with 1σ uncertainty and correlation matrix	e.g. hull C_d 0.402 ± 0.9 % (synthetic data)



Where the energy goes (EnergyBudget scenario: 60 s at 1 Hz, 1.8 W). With placeholder parameters the motor is oversized, so its winding dominates. In SwimForward only ≈ 0.01 % of battery energy becomes useful propulsion work. The demo commits no plots; they are generated by the scripts.

Demo stages

- 1–2 Pipe, chamber and relief valve with tests (laminar, turbulent vs MSL, closed loop)
- 3–4 Battery, H-bridge, motor, pump; tail, CPG and the TailDrive subsystem; frequency sweep
- 5–7 Energy budget; ballast and depth control; fin and surge, forward swimming
- 8–9 FMU export and FMPy loop; Modelica.Fluid variant (same results within 0.5 %, 6× slower)
- 10 Calibration benches and calibrate.py for 10 subsystems

Limitations

- The tail has one degree of freedom (no travelling wave); thrust is a placeholder Lighthill formula.
- Surge, heave and yaw are not coupled; drag and added mass are constants.
- Lumped incompressible hydraulics without hysteresis; ideal electrics. The FMU runs only on the OS it was built on.

4 · Comparison

Capability matrix

● modelled in the demo ◐ simplified or partial ○ not modelled

CAPABILITY	MUJOCO	STONEFISH	PYELASTICA	SOFA	OPENFOAM FSI	OPENMODELICA
Free swimming of the whole fish	●	●	●	○ clamped	○ head fixed	◐ surge + heave
Soft tail deformation	◐ 5 segments	◐ 5 segments	◐ continuous rod	● 3D FEM	● 2D FEM	○ 1 DOF
Chamber pressure from structure	◐ constant C_h	◐ constant C_h	◐ constant C_h	●	◐ pressure is input	◐ p-V table
Water flow, wake, vortices	○	○	○	○	●	○
Hydrodynamic forces	◐ ellipsoid model	◐ per mesh face	◐ Morison + Lighthill	◐ drag only	● Navier–Stokes	◐ Lighthill + C_d
Added mass	◐ diagonal	◐ axis-averaged	◐ trailing edge	○	● from the flow	◐ constant
Pump and valve dynamics	◐	◐	◐	◐	○	●
Electrics and energy budget	○	○	○	○	○	●
Buoyancy and depth control	●	●	●	○	○	◐ 1D heave
Noisy sensors	○ possible	●	○	○	○	○
Currents and environment	○	●	○	○	◐ inflow	○
Interactive 3D view	●	●	○ GIF	●	◐ ParaView, after the run	○ plots
Real time or faster	●	● 5–8×	◐ ≈1.5× slower	○ 100–800× slower	○ hours	● milliseconds
Parameter identification	○ plan only	○	○ plan only	○ plan only	○	● calibrate.py
Co-simulation / export	○	◐ ROS 2 possible	○	○	● preCICE	● FMU 2.0

Inputs at a glance

INPUT	 MUJOCO	 STONEFISH	 PYELASTICA	 SOFA	 OPENFOAM FSI	 OPENMODELICA
Geometry	MJCF generated in Python	OBJ meshes + .scn XML	station table in Python	gmsh → VTK + OBJ	gmsh → OpenFOAM mesh + CalculiX mesh	none (lumped parameters)
Material	densities, joint stiffness	densities, joint stiffness	E, v, p per element	E, v, p, fibers, spine	E, v, p	tail k, J, c
Hydraulics	lumped (A, r, C_h)	lumped (A, r, C_h)	lumped (A_r, C_h)	volume per chamber	pressure per chamber	component parameters + p–V table
Water	fluidcoef	from the mesh + lift slope	C_n, C_t, model choice	C_n, C_t	ρ, v, inflow U	C_d, C_T, added mass
Config format	Python dataclass	JSON + XML	Python dataclass	Python dataclass + env vars	dictionaries, .inp, XML	Modelica parameters
Live control	keyboard	keyboard	none	GUI	none	none

Outputs at a glance

OUTPUT	 MUJOCO	 STONEFISH	 PYELASTICA	 SOFA	 OPENFOAM FSI	 OPENMODELICA
Main quantity	speed, trajectory, depth	speed, trajectory, sensor readings	speed, body shape, thrust	pressure, bend, tethered thrust	flow field, forces, wake	currents, pressures, energy
Data format	CSV	CSV 100 Hz + JSON	CSV	CSV + NPZ	OpenFOAM fields, .frd, logs	.mat, CSV
Visual	viewer, PNG	3D GUI, PNG	PNG, GIF	GUI, MP4	ParaView, PNG	PNG plots

Same fish, different speeds: why

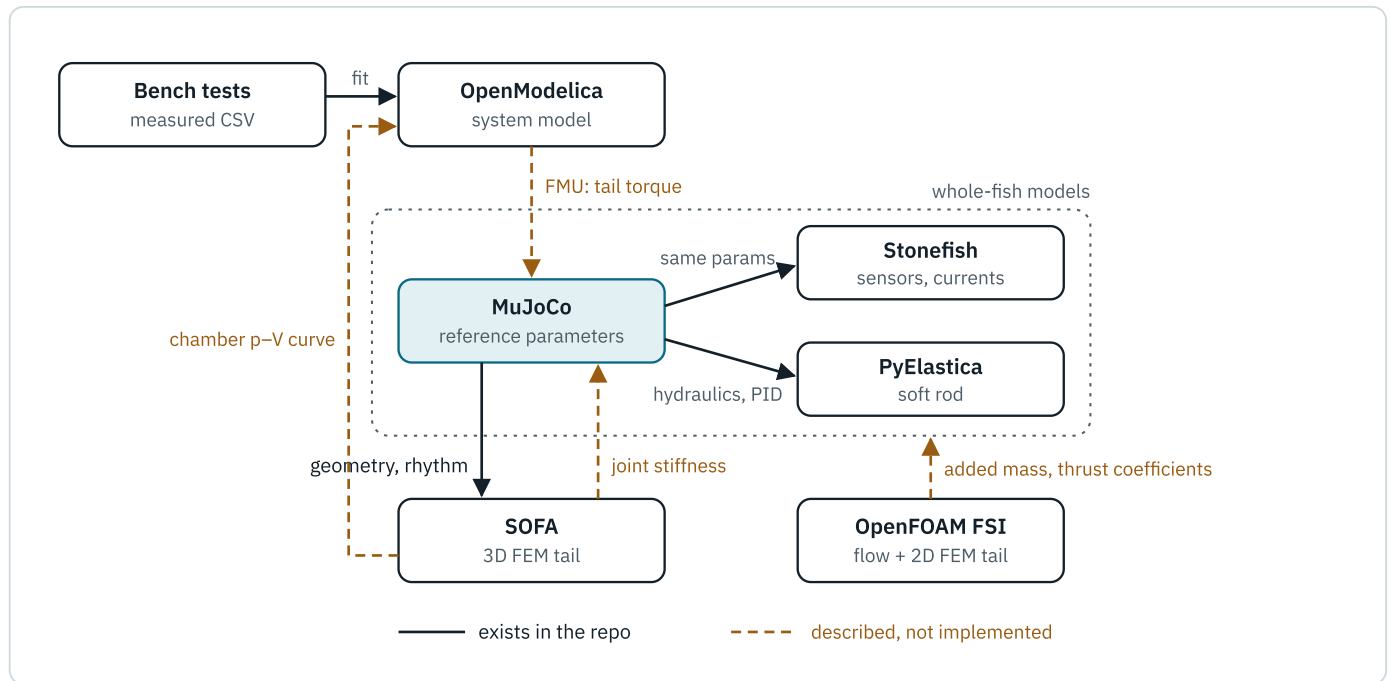
At about 2 Hz the three free-swimming demos give speeds that differ by 2×. The difference comes from where each tool gets its thrust.

TOOL	SPEED NEAR 2 HZ	MAIN SOURCE OF THRUST IN THE MODEL
 MuJoCo	22 cm/s	Kutta lift on ellipsoids from the built-in model
 Stonefish	34 cm/s	flat-plate fin lift added in C++
 PyElastica	44 cm/s	Lighthill reactive force at the trailing edge
 OpenModelica	7.3 cm/s	Lighthill rigid fin on a 1-DOF tail (only 3.4° amplitude at 2 Hz)
 SOFA	tethered	drag on skin triangles: 63 mN at 2 Hz
 OpenFOAM FSI	tethered	resolved flow: net drag with this tail shape

Cost of use

TOOL	INSTALL	TYPICAL RUN	SKILLS NEEDED
 MuJoCo	pip, minutes	seconds to minutes	Python
 Stonefish	build from source, GPU for the GUI	1–2 min for all scenarios	C++, XML
 PyElastica	pip, minutes	9–70 s per stage	Python, rod mechanics
 SOFA	script, ≈2 min + compiler for CHOLMOD	10 min to 1 h per stage	Python, FEM, meshing
 OpenFOAM FSI	≈1.5 h, three programs and adapters from source	7–102 min per case	CFD, FEM, coupling
 OpenModelica	apt repository + venv	under a second	Modelica

How the tools fit together



MuJoCo is the reference: Stonefish reuses its fish and parameters, PyElastica ports its hydraulics and depth PID, and SOFA copies its tail geometry and rhythm. The dashed links are the planned next steps, each described in a README: SOFA p-V curves into the OpenModelica chamber table, the OpenModelica FMU driving a MuJoCo joint, SOFA/PyElastica stiffness exported to MuJoCo joints, and OpenFOAM results used to calibrate the cheaper water models.

5 · Glossary

ENGLISH	POLISH	MEANING HERE
added mass	masa dodana	water that has to be accelerated together with the body; in the FSI demo it lowers the tail resonance from 3.39 Hz to ≈ 1 Hz
CPG (rhythm generator)	generator rytmu	controller that produces the periodic volume command for the tail
Cosserat rod	pręt Cosseratów	1D continuum that bends, twists, shears and stretches (PyElastica)
corotational FEM	MES korotacyjna	linear elasticity that handles large rotations (SOFA)
FMU	FMU (Functional Mock-up Unit)	packaged model that other tools can run (OpenModelica export)
FSI	interakcja płyn–struktura	coupled simulation of water flow and a deforming solid (OpenFOAM + CalculiX)
IQN-ILS	IQN-ILS (quasi-Newton)	method preCICE uses to make implicit coupling converge
Lighthill reactive force	siła reakcyjna Lighthilla	thrust from the water pushed sideways by the trailing edge
p–V curve	krzywa ciśnienie–objętość	chamber pressure as a function of the water volume pumped in
PRBM	model pseudo-sztywny	soft tail approximated by rigid segments and springs (MuJoCo, Stonefish)
relief valve	zawór przelewowy	valve that opens at $ \Delta p = 50$ kPa to protect the chambers
Strouhal number	liczba Strouhala	$St = f \cdot A / U$; efficient fish swim at $St \approx 0.2$ – 0.4
tethered	na uwięzi	body held in place while thrust is measured (SOFA, OpenFOAM)
VBS	układ zmiennej pływalności	variable buoyancy system: a tank whose water volume sets depth (Stonefish)

* Licences are not stated in the repository. The licences listed are those of the tools themselves, from their public distributions.